On the Guidance of Flow Matching: Review on synthetic data and towards cell data

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Motivation & Core Objective

- **Flow Matching (FM):** Generative method learning a velocity field $v_t(x)$ to transport a tractable source distribution p_0 into a complex data distribution p_1.
- **The Question:** How can we generate *specific parts* of the complex target distribution?
- **Solution:** Energy-based guidance! Steer a frozen model toward a defined objective.

1. Notation & Abbreviations Overview

- **Variables & Distributions:**
  - $p_0 \to p_1$: Source distribution to Target data distribution
  - $v_t(x)$: Learned base velocity field
  - $J(x)$: Energy function (defines the target subset)
  - $g_t(x_t)$: Guidance vector field (proportional to $-\nabla J$)
  - $v' = v_t + g_t$: Guided velocity field
  - $z = (x_0, x_1)$: Coupling of source and target
  - $\pi(x_0, x_1)$: distribution over the source-target coupling
  - $\mathcal{P}$: Reverse coupling ratio ($\mathcal{P} = 1$ for independent coupling)
- **Guidance Methods:**
  - **MC** (g^{MC}): Monte Carlo estimation of the intractable integral (due to $p(z|x_t)$) describing the guidance vector field.
  - **sim-MC** (g^{sim-MC}): Simulated Monte Carlo Gradient Guidance. Approximates the intractable distribution with a Gaussian. Samples from this simpler shape to estimate guidance when the energy function lacks a clean math formula.
  - **cov-A** (g^{cov-A}): Affine path covariance approximation. Uses Taylor expansion to approximate guidance locally around the mean.
  - **cov-G** (g^{cov-G}): Affine path covariance approximation. Assumes a standard Gaussian path to mathematically map flow matching back to classic diffusion guidance. Computes exact covariance using network gradients, offering strong performance at a higher computational cost.
  - **GM / Learned** (g_ϕ): Guidance Matching. Training-based method where a neural network is explicitly trained to become the guidance vector field. Trains multiple models: one for Z to estimate the normalizing constant, and a second model that directly learns the guidance vector field.

2. Steering Mechanisms

- **Training-time Steering (Conditioning):** Requires data and a label y during training. The model reproduces seen labels directly.
- **Inference-time Steering (Guidance):** Train the model purely on data. Keep the base flow model frozen. Add guidance g_t at sampling to push the flow toward low energy J.

3. Guidance

- **Guided Target Distribution:** Reweighting the original target $p_1(x)$ by the energy J.

$$p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$$

4. The Hidden Assumption & Phase Diagram

- Guidance validity relies heavily on the reverse coupling ratio: $\mathcal{P} = \frac{\pi'(x_0|x_1)}{\pi(x_0|x_1)}$.
- **When it works:** Valid when $\mathcal{P} \approx 1$ (have $\mathcal{P} = 1$ for independent coupling).
- **When it breaks:** The guidance g_t points in a systematically wrong direction when coupling is **strong** (e.g., Exact Optimal Transport) AND the energy J is **sharp**.

5. Comparison of Coupling Methods

Feature	**Independent**	**Entropic OT**	**Exact OT**
Mapping Style	All-to-All (Random)	"Fuzzy" 1-to-Many	1-to-1 (Deterministic)
Path Trajectories	Highly crossing, complex	Localized, less crossing	Straight, direct, non-crossing
VF Smoothness	Very High	Moderate to High	Low (Sharp)